MODULE 4 PREGNANCY AND BIRTH

Pregnancy and Birth.

PREGNANCY AND BIRTH

A reference for the pregnancy and birth video and lab. This page covers fertilization and implantation, the stages of prenatal development, the placenta and fetal membranes, the gravid uterus, and the stages of labor.

BY THE END

1. Describe fertilization and implantation.
2. Order the stages of prenatal development from zygote to fetus.
3. Name the placenta and the fetal membranes and what each does.
4. Describe the gravid uterus and the three stages of labor.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Pregnancy and birth

drsrennie-stack.github.io/new-build-bio4-solano/pregnancy-concept-videos.html

Pregnancy and Birth, an Overview

Pregnancy begins at fertilization and ends at birth. Over those months a single cell becomes a fetus, supported by a temporary set of structures built just for the pregnancy.

THE VOCABULARY OF A PREGNANCY

Structure	What it is
Pregnancy	The period from fertilization to birth, during which a fetus develops in the uterus
Gestation	The length of a pregnancy, about 38 weeks from fertilization, or 40 weeks counted from the last menstrual period
Embryo	The developing offspring during the first 8 weeks after fertilization
Fetus	The developing offspring from the 9th week until birth
Trimesters	The three roughly three-month periods that pregnancy is divided into

Fertilization and Implantation

Pregnancy begins when a sperm and an egg unite. The new cell divides as it travels, and about a week later it embeds itself in the wall of the uterus.

FROM ONE CELL TO AN IMPLANTED BLASTOCYST

Structure	What it is
Fertilization	The union of a sperm and an egg, normally in the ampulla of the uterine tube
Zygote	The single cell formed at fertilization
Cleavage	The series of rapid cell divisions the zygote undergoes as it travels toward the uterus
Morula	The solid ball of cells produced by cleavage
Blastocyst	The hollow ball of cells that reaches the uterus, with an inner cell mass and an outer layer
Inner cell mass	The cluster of cells of the blastocyst that becomes the embryo
Trophoblast	The outer layer of the blastocyst that becomes the fetal part of the placenta
Implantation	The embedding of the blastocyst into the endometrium, about a week after fertilization

HOLD ONTO THIS

Fertilization happens in the ampulla, implantation happens in the endometrium, and the week between them is spent travelling and dividing. Two places, one journey.

Stages of Prenatal Development

From one cell to a newborn, development passes through a fixed set of stages. Follow them in order.

1. **Zygote** the single cell formed when a sperm and an egg unite
2. **Cleavage** rapid cell divisions produce a solid ball of cells, the morula
3. **Blastocyst** a hollow ball of cells that reaches the uterus and implants in the endometrium
4. **Embryo** from implantation through week 8; the organs and body systems form
5. **Fetus** from week 9 until birth; the body grows and the systems mature
6. **Birth** the fetus is delivered, ending the gestation

The Placenta and Fetal Membranes

A pregnancy builds temporary structures that support the fetus and are discarded at birth. Compare the placenta and the fetal membranes.

THE TEMPORARY STRUCTURES OF PREGNANCY

Structure	What it is	Role
Placenta	The disc-shaped organ that forms during pregnancy	Exchanges oxygen, nutrients, and wastes between the mother's blood and the fetus's blood
Umbilical cord	The cord linking the fetus to the placenta	Carries the two umbilical arteries and the umbilical vein
Amnion	The innermost fetal membrane	Forms the fluid-filled sac that surrounds the fetus
Amniotic fluid	The fluid within the amnion	Cushions the fetus, lets it move, and allows even growth
Chorion	The outermost fetal membrane	Contributes the fetal portion of the placenta
Yolk sac	An early fetal membrane	Forms the earliest blood cells and the germ cells

The Gravid Uterus and Fetal Position

The uterus changes more than any other organ during pregnancy, and the fetus settles into position for birth.

THE PREGNANT UTERUS AND HOW THE FETUS LIES

Structure	What it is
Gravid uterus	The uterus during pregnancy, which enlarges enormously to hold the growing fetus
The expanding uterus	By full term it reaches up to the rib cage, pushing the abdominal organs aside
Cervical changes	The cervix stays firm and closed through pregnancy, then softens and thins as birth approaches
Vertex position	The head-down position most fetuses settle into by late pregnancy, ready for birth
Breech position	A position in which the buttocks or feet, rather than the head, are nearest the cervix
Lightening	The descent of the fetus lower into the pelvis as birth nears

Labor and Birth

Labor is the process of birth. It is divided into three stages, which follow in a fixed order.

1. **Dilation stage** regular contractions thin and open the cervix until it is fully dilated; the longest stage
2. **Expulsion stage** strong contractions push the fetus through the cervix and vagina; it ends with the birth of the baby
3. **Placental stage** contractions continue and expel the placenta, now called the afterbirth

THE VOCABULARY OF BIRTH

Structure	What it is
Labor	The process by which the fetus and placenta are expelled from the uterus
Afterbirth	The placenta and attached membranes, expelled in the placental stage
Cesarean section	Surgical delivery of the fetus through an incision in the abdominal and uterine walls

Disorders of Pregnancy

Compare the common conditions of pregnancy by the structure or stage each one affects.

THE COMMON CONDITIONS OF PREGNANCY

Condition	Affected structure or stage	What it is
Ectopic pregnancy	The implantation site	The embryo implants outside the uterus, most often in a uterine tube; a structural emergency
Miscarriage	The pregnancy	The loss of a pregnancy before the fetus can survive on its own
Placenta previa	The placenta	The placenta lies over the cervical opening and can block the birth canal
Preeclampsia	The mother's blood pressure	A serious condition of later pregnancy marked by high blood pressure
Gestational diabetes	Blood sugar control	High blood sugar that first develops during pregnancy
Breech presentation	Fetal position	The fetus is positioned buttocks or feet first, which can complicate a vaginal delivery

TEMPORARY STRUCTURES, LASTING CLUES

The **placenta** is the exchange organ of pregnancy, and where it sits matters. In **placenta previa** it lies over the cervix, blocking the path the baby would take, which is why position is checked on ultrasound and may call for a cesarean section.

The **amniotic sac** normally stays sealed until labor. Its rupture, felt as the water breaking, releases the amniotic fluid and signals that birth is near. A sample of that fluid can also be drawn, in amniocentesis, to study the fetus.

The fetal membranes and placenta are built for one pregnancy and discarded at birth as the **afterbirth**. Examining the delivered placenta afterward gives clues to how the pregnancy went, a temporary organ read like a record.